Question 1. a) Explain the essential technologies in full stack development with an architecture?

Answer 1. Full stack development involves building both frontend and backend of a web application.

Essential technologies:

1) Frontend:
HTML, CSS, Javascript. — for structure, styling and interactivity
* Frameworks: React, Angular, vue.js.

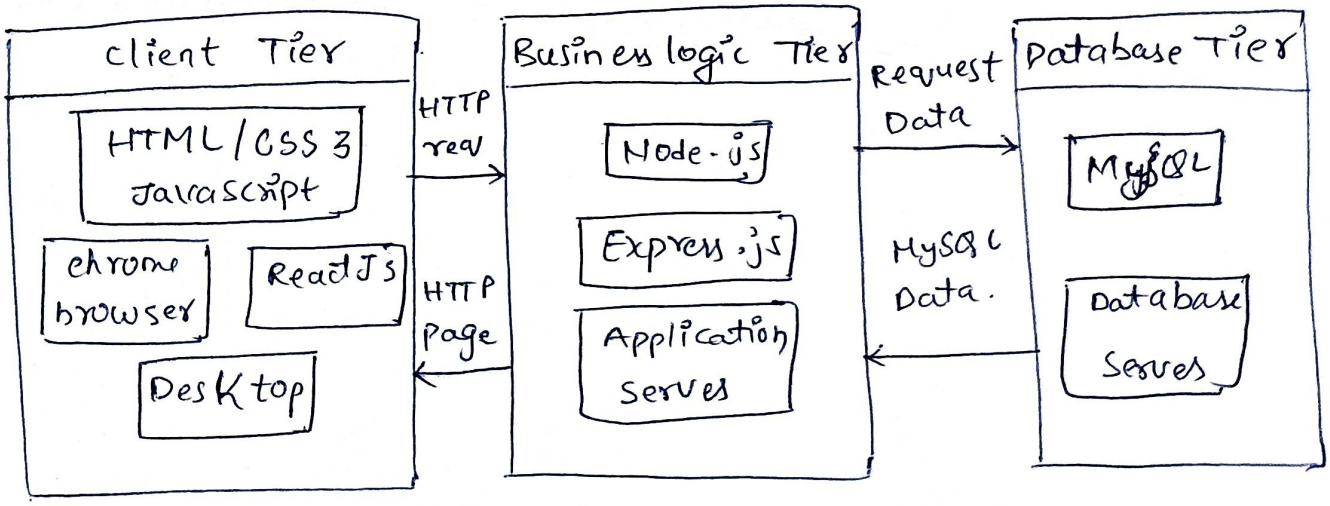
2) Backend:
* languages / frameworks - Node.js, Express.js, Django, spring boot
* handles business logic and API creation.

3) Database:
* SQL or NO SQL.

4) Version control:
* Git, Github - for tracking code changes

5) Deployment tools:
* Docker, AWS, Heroku, Vercel - for hosting and deployment.

Architecture Diagram:



Question 2. B) Compare ORM tools and Hibernate.

Answer 2: ORM (Object Relational Mapping :-

Object Relational Mapping (ORM) is a programming technique that bridges the gap between object-oriented programming languages and relational databases. It allows developers to work with data as objects rather than tables, eliminating the need to write complex SQL queries manually. ORM frameworks automatically handle the mapping between classes and database tables, making database operations easier and reducing repetitive code.

Hibernate :-

Hibernate, on the other hand, is a specific implementation of the ORM concept designed for Java applications. It provides a robust framework for managing database connectivity, query execution, and object persistence. Unlike general ORM tools, Hibernate offers advanced features such as automatic schema generation, caching, lazy loading, and transaction management.